Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	Indomie
Selected Domain	Domain 1
Selected SDG Track (1–17)	1, 4, 8, 9, 11
Project / Solution Name	Nordyc
Name (Leader)	Nelsen Keane Gunawan
Email & Phone (Leader)	DMT2309281@xmu.edu.my & (+62) 859 2065 5352
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is reshaping work and the skills required across industries. According to the World Economic Forum's *Future of Jobs Report 2025*, employers expect 39% of workers' core skills to change by 2030, driven partly by technological advancements and AI. In Malaysia, the OECD reported that 35.6% of tertiary graduates experienced skill-related underemployment in 2025, indicating an existing mismatch between graduates' qualifications and labour-market needs. This highlights the need for universities to prepare graduates for changing industry demands.

However, university career centres often lack a structured way to assess whether existing courses remain relevant to industry needs. Curriculum reviews occur periodically, require staff to manually examine many programmes and courses, and detailed syllabi are often unavailable. Without up-to-date, dated evidence, decisions can go wrong in two directions: universities may overlook emerging skill gaps, or wrongly treat valuable foundational courses as obsolete simply because AI can perform some related tasks. The consequences extend beyond education. Economically, a mismatch between graduates' skills and industry needs can affect employability and productivity. Socially, graduates may struggle to access suitable jobs, while universities risk losing relevance.

This challenge relates to SDG 4.4, which promotes skills for employment, and SDG 8.5 and 8.6, which focus on decent work and youth employment, education, and training. Universities therefore need a transparent, repeatable, and evidence-based approach to identify courses requiring academic review and recommend the least disruptive changes, while leaving final decisions to academic staff.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

We propose **NORDYC**, an AI-powered University Curriculum and Career Intelligence Platform designed to help universities align their academic programmes with evolving industry demands. The system will take a university major's study plan as input and analyse existing curricula, industry skill requirements,

software platforms, course durations, labour-market data, and AI developments to generate evidence-based recommendations for curriculum improvement.

NORDYC will collect job-posting data, labour-market reports, and AI-related news or research to identify emerging occupations, declining opportunities, and changing skill requirements, alongside extracting relevant competencies, technical tools, and qualifications from these sources. NORDYC will compare these findings with existing modules to identify skill gaps and assess the relevance of individual courses and entire academic programmes.

Based on the analysis, the platform will generate several outputs. *Intake specific learning plan* will prioritize skills to learn according to students academic stages and remaining study duration. *Refined syllabi and elective recommendations* will identify content to introduce, update, or remove based on potential career pathways for each major. *A programme relevance overview and course relevance overview* will help universities evaluate their curriculum alignment with industry requirements. *A future workspace radar* will visualize changing job demand, while alternative AI development scenarios will help universities prepare for different levels of technological disruption. Finally, an *academic action plan* will outline recommended changes, implementation priorities, and timelines

NORDYC will use transparent scoring methods to evaluate skill demand and curriculum gaps, supported by a Generative AI model that explains findings and drafts recommendations grounded in a collected database. Rather than guaranteeing AI-proof occupations, it will highlight careers with comparatively resilient prospects. Relevant careers and skills will also be mapped to the United Nations Sustainable Development Goals (SDGs). Ultimately, the platform will help universities make informed curriculum decisions, plan improvements for different student intakes, and prepare graduates for evolving employment opportunities.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend	React, TypeScript, Vite, React Router	Provides the programme library, curriculum uploads, course intelligence, recommendations, pathways, scenarios, and Action Plan dashboard.
Backend/API	Python, FastAPI, Uvicorn	Exposes protected APIs for programme records, document ingestion, analysis jobs, evidence retrieval, and human review.
Validation	Pydantic, JSON Schema, AJV	Validates API data and structured AI outputs. Additional checks verify course identities, evidence references, and citations.
Database	PostgreSQL, SQLAlchemy, Psycopg, Alembic	Stores versioned curricula, evidence, analysis results, job states, Action Plans, and review history. Alembic manages schema migrations.
Document processing	pdfplumber, bounded extraction subprocesses	Extracts curriculum information from supported PDFs and flags ambiguous layouts for review. Reviewed JSON import provides an alternative.
Background processing	Separate Python worker with database-backed jobs	Processes document extraction and analysis without blocking interactive API requests.
Generative AI	Anthropic Claude through Replit-managed AI	Produces structured, evidence-grounded findings through the existing Claude adapter. The current synthetic demo substitutes explicitly mocked responses.

External research	Brave Search API, HTTPX, Beautiful Soup	Supports controlled discovery of relevant sources. Search results are not automatically accepted as verified evidence.
Evidence sources	Reviewed ESCO, MQA, workforce, AI-capability, and UN SDG references	Supplies traceable benchmarks and dated signals, subject to source permissions, coverage, and currentness limits.
Development and hosting	GitHub, Replit	GitHub maintains source and CI workflows. Replit currently runs the controlled development preview; public deployment remains a separate step.
Testing	pytest, Vitest, Playwright, Axe	Checks backend behaviour, frontend components, browser workflows, and selected accessibility requirements.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

- Empathise: We centred the design on university career centres and academic committees, recognising that detailed syllabi may be unavailable and academic decisions require traceable evidence.
- Define: We framed the challenge as identifying curriculum review priorities—not automatically declaring courses obsolete or replacing human judgement.
- Ideate: We developed an upload-first workflow, evidence-linked findings, and alternative interventions, including syllabus revision, electives, industry engagement, and no change.
- Prototype: We implemented a React dashboard connected to FastAPI and PostgreSQL. Synthetic programmes support safe demonstrations, while asynchronous processing keeps document extraction and analysis separate from interactive requests.
- Test: Backend, frontend, and browser checks assess technical behaviour, permissions, accessibility, and error handling. These checks establish software functionality, not academic accuracy; structured user evaluation remains necessary.

For competition Phase 2, deployment will progress from the controlled prototype to a restricted institutional pilot. Priorities include named-user authentication, institution-scoped access, persistent private document storage, supervised workers, backups, monitoring, and consent-controlled AI processing. Pilot users will evaluate task completion, clarity, evidence usefulness, and review workflows before wider

release. This staged approach supports technical viability and usability without overstating production readiness.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	● 4.4: "Increase the number of people with relevant skills for financial success" ● 8.5: "Full employment and decent work with equal pay" ● 8.6: "Promote employment, education, and training"
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	● According to the World Economic Forum's *Future of Jobs Report 2025*, employers expect 39% of workers' core skills to change by 2030, driven partly by technological advancements and AI. ● In Malaysia, the OECD reported that 35.6% of tertiary graduates experienced skill-related underemployment in 2025, indicating an existing mismatch between graduates' qualifications and labour-market needs. ● Together, these findings highlight the challenge of keeping university curricula aligned with rapidly evolving industry requirements.
Core Measurable KPI *The specific unit you will track*	● Industry-Relevant Skills Coverage (%): Percentage of high-priority skills identified from current job

Metric Category	Your Response
e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.	postings that are adequately covered by a university major's curriculum. ● Employment Rate (%): Percentage of graduates employed within a defined period after graduation. ● Layoff Rate (%): Percentage of employees laid off within a defined period, potentially tracked by occupation or industry where reliable data is available.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	● Improve the user interface: Redesign NORDYC's dashboards, visualisations, and reports to make curriculum relevance, industry skill gaps, and workforce trends easier for university administrators to understand and act upon. ● Enhance curriculum planning features: Refine intake-specific learning plans and AI-disruption scenarios to help universities plan curriculum changes according to student cohorts' academic stages and different possible futures. ● Prepare for wider adoption: Design a scalable data structure and analysis workflow that can accommodate more majors and universities across Malaysia, with the long-term goal of supporting universities internationally.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [v] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- [v] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- [v] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- [v] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
	Dataset	

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Nelsen Keane Gunawan

24 October 2026